

Pilot Study Design

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To validate the effectiveness of the **modular HDPE dome system** under real-world conditions, a **controlled pilot study** will be conducted. This study is designed to assess ecological safety, habitat enhancement, and practical feasibility before scaling up to larger mangrove areas.

Location

The pilot will take place on a **50 m² degraded mangrove patch** within the Esmeraldas–Pacific Colombia Mangroves. This patch has been selected because it is representative of vulnerable habitats, with weakened roots, shallow sediment, and limited juvenile marine life presence, making it ideal for testing the dome system's impact.

Features: Rounded edges, escape gaps, UV stabilization, modular dome geometry

Duration

The study will run for **6–12 months**, covering a **full seasonal cycle** to capture:

- Natural **mangrove growth patterns** during both wet and dry seasons.
- Seasonal fluctuations in **juvenile shark and fish activity**, sediment dynamics, and water quality.
- The dome's performance during periods of **high rainfall, wave action, and sediment movement**.

Key Metrics to Monitor

1. Mangrove Root Growth Rate:

- Track the **establishment, vertical growth, and structural strength** of seedlings.
- Assess how the domes aid **root anchoring and early survival** against wave and sediment stress.

2. Sediment Depth and Stability:

- Measure **changes in sediment accumulation or erosion** around the dome.

- Determine the effectiveness of the module in **preventing sediment loss and stabilizing the substrate**.

3. **Juvenile Shark and Fish Sightings:**

- Monitor **habitat use, abundance, and behavior** under the domes.
- Evaluate whether the structures provide **safe micro-habitats** that improve early-stage survival rates.

4. **Water Quality and Toxicity Levels:**

- Conduct regular **chemical and physical water tests** to ensure HDPE domes remain inert and safe.
- Test for pH, salinity, and potential leachates to confirm **no harmful impact on marine life**.

Purpose and Expected Outcomes

This pilot study will generate **measurable, repeatable data** to:

- Demonstrate the **ecological effectiveness** of the dome system in protecting mangrove roots and supporting juvenile marine life.
- Verify the **chemical and physical safety** of the HDPE materials in a real-world setting.
- Provide a **practical blueprint for scaling** the project to larger mangrove restoration areas, informing **deployment strategies, monitoring protocols, and resource allocation** for long-term conservation efforts.

By the end of the pilot, researchers and stakeholders will have **actionable insights** into both ecological and operational aspects of the dome system, ensuring that the full-scale project is scientifically grounded, safe, and impactful.